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Original Article

Bilateral PECS II Block is Associated with Decreased Opioid Consumption and Reduced Pain Scores for up to 24 hours After Minimally Invasive Repair of Pectus Excavatum (Nuss Procedure): A Retrospective Analysis[☆]

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ABSTRACT

Objective(s): Compared to the open surgical technique, the minimally invasive repair of pectus excavatum (MIRPE; Nuss procedure) is a thoracoscopic technique designed to minimize intraoperative tissue damage. It still causes severe postoperative pain due to the insertion and pressure of the retrosternal bar used to raise the sternum and stabilize the chest. This study aimed to identify associations between ultrasound-guided PECS-II block and postoperative analgesia after the Nuss procedure.

Design: A retrospective cohort study

Setting: Single-center, training and research hospital affiliated with a university

Participants: From Jan 1, 2018 to Nov 15, 2021, 171 consecutive patients were identified who underwent MIRPE surgery under general anesthesia. All patients received intravenous (I) patient-controlled analgesia (PCA) with or without PECS-II blocks for postoperative analgesia. One hundred twenty-five patients who met the inclusion criteria were evaluated.

Interventions: Demand-only morphine intravenous PCA was used for postoperative pain management in the PECS and control groups. Bilateral PECS-II block with 0.25% bupivacaine was performed in to the PECS group.

Measurements and main results: The primary outcome was postoperative opioid consumption, calculated as mg/kg of IV morphine. Secondary outcomes included Numeric Rating Scale (NRS) pain scores at rest (static) and with movement (dynamic) recorded 1, 4, 8, 12, 24 h after surgery. Postoperative morphine consumption was significantly lower in the PECS group than in the control group over the first 24 hours postoperatively: 0.325 mg/kg vs. 0.425 mg/kg ($p < 0.001$). Static and dynamic NRS values were significantly lower in the PECS group for the first 12 postoperative hours ($p < 0.05$).

Conclusions: Bilateral PECS-II block is associated with decreased pain scores for up to 12 hours, and with decreased opioid consumption for up to 24 hours, following minimally invasive repair of pectus excavatum (Nuss procedure) in adolescents. PECS-II block in this context has not been previously described.

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Abbreviations: MIRPE; Minimally invasive repair of pectus excavatum

[☆]The study entitled “Bilateral PECS II block is associated with decreased opioid consumption and reduced pain scores for up to 24 hours after minimally invasive repair of pectus excavatum (Nuss procedure): a retrospective analysis” was performed in Bakirkoy Dr. Sadi Konuk Education and Research Hospital, Department of Anesthesiology and Reanimation, Istanbul, Turkey.

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Keywords: Pectoral nerve block; MIRPE; Nuss procedure; postoperative analgesia; pectus excavatum; numerical rating scale

Introduction

Pectus excavatum (PE) is the most common congenital chest wall deformity in children and young adults, characterized by symmetrical or asymmetrical chest depression and sternocostal deformation. Pectus excavatum commonly is related to other connective tissue dysplastic diseases and affects one out of every 400 live births.^{1,2} Males are more likely to have a chest wall condition called PE than females.³ Depending on the severity of the deformity, problems associated with pectus excavatum range from cosmetic deformity to serious cardiopulmonary symptoms, such as shortness of breath and exercise intolerance,⁴ that commonly are considered indications for surgical treatment.^{5,6}

The traditional open surgical technique (Ravitch) and the thoracoscopic minimally invasive technique are the two main surgical techniques used to repair PE. Minimally invasive repair of pectus excavatum (MIRPE) (Nuss procedure) is more advantageous than the traditional open technique in terms of the shorter surgical time, minimal intraoperative blood loss, outstanding cosmetic results, lower incidence of complications, and decreased length of hospital stay.⁷⁻⁹

In addition, it frequently has become preferred for PE patients due to the possibility of more visible surgical scarring after the open technique.¹⁰ Although the MIRPE is a thoracoscopic technique that minimizes tissue damage, it may cause severe postoperative pain due to the pressure applied by the bars placed behind the anterior chest wall.¹⁰ Recently, “Enhanced Recovery After Surgery (ERAS)” programs, including guidelines for thoracic surgery, have recommended thoracic epidural and paravertebral block as the gold standard method for postoperative pain management in patients undergoing thoracic surgery.¹¹ Although the complication rate of these techniques is much lower in the pediatric age group than in adults, serious potential complications, such as intrathecal spread, epidural hematoma, epidural abscess and pneumothorax, cannot be ignored.¹²

Regardless of the type of analgesic treatments, most patients report moderate-to-severe pain following the Nuss procedure.¹³ Anesthesiologists continuously look for innovative ways to promote efficient analgesia while minimizing side effects and complications. Interfascial plane blocks, such as the erector spinae plane block (ESPB), serratus anterior plane block (SAPB), and pectoral nerve block (PECS), recently have gained popularity for postoperative regional analgesia.

PECS-II is an interfascial plane block in which local anesthetic is injected between the pectoralis major, pectoralis minor, and serratus anterior muscles. PECS-II is a less invasive and easier to perform alternative to thoracic paravertebral block (TPVB). This block, defined by Blanco et al., provides adequate and safe postoperative analgesia in the anterior chest wall after breast surgery.^{14,15} However, its use in thoracic

surgery is limited.¹⁶ There are no current data in the literature assessing PECS-II block in the context of the MIRPE.

This study aimed to identify associations between PECS-II block and analgesic outcomes following minimally invasive repair of pectus excavatum, specifically postoperative opioid consumption (primary outcome) and postoperative NRS pain scores, at rest and with movement (static and dynamic; secondary outcomes).

Material and methods

Study design

After obtaining approval from the Clinical Research Ethics Committee of the training and research hospital where the study was conducted (2021-503/2021-20-10), patients older than 14 and with the American Society of Anesthesiologists (ASA) physical status I-II who underwent Nuss procedure were included to the study.

In 2018, the authors' department started performing PECS block on this patient population. Therefore, they chose the historical consecutive data to begin in January 2018. From Jan 1, 2018 through Nov 15, 2021, one hundred seventy-one consecutive patients were identified who had an MIRPE surgery with or without PECS-II block in addition to postoperative demand-only morphine IV PCA. Ninety-seven patients were identified who did not receive PECS-II blocks; seventy-four patients were identified who did receive PECS-II blocks. If the patient's PECS-II block note did not expressly state the use of IV PCA, the patient was excluded. In addition, patients who received PECS-II block with other than 0.25% bupivacaine were excluded. Also, patients having ASA physical status classification $\geq$ III, incomplete data, and underwent other interfascial plane blocks were excluded from the study. One hundred twenty-five patients who met the inclusion criteria were analyzed according to the analgesic method they received. Institutional computer-based documentation systems and patient and pain follow-up assessment forms were used to collect data. The study was conducted by the principles stated in the Declaration of Helsinki (as revised in 2013).

General anesthesia technique

Before surgery, 22-gauge vascular access was established in all patients, and infusion with 2-4 mL/kg/h Ringer's lactate solution was started. Standard monitoring was achieved with electrocardiography (ECG), noninvasive blood pressure, and peripheral oxygen saturation (SpO₂). After monitoring, 0.03-0.05 mg/kg of midazolam were administered as premedication. After induction was achieved with 1.5 μ g/kg of fentanyl, 2-3 mg/kg of propofol, and 0.6-0.8 mg/kg of IV rocuronium, the patients were intubated with a double-lumen tube. Anesthesia

was maintained with 1.5-2% sevoflurane in 40% oxygen in air, with a total inflow of 2.5 L/min and iv remifentanyl infusion (0.05-0.1 $\mu\text{g/kg/min}$). In addition, all patients were administered 15 mg/kg IV paracetamol at the beginning of the surgery, and 4 mg of ondansetron to prevent nausea and vomiting at the end of the surgery. At the end of the surgery, all patients were followed routinely in the recovery room with mask O_2 (2 L/min) for 15 minutes, and then they were sent to the post-anesthesia care unit (PACU).

PECS-II block technique

PECS-II block was performed under ultrasound guidance with the patients in the supine position and arms in abduction, following anesthesia induction and endotracheal intubation. A single anesthesiologist (GS) with more than five years of experience performed all PECS-II blocks. Blocks were performed using a 50-mm 22-gauge needle (Stimupleks Ultra 360 30° - BRA-04892510-01 / B. Braun Melsungen AG, Japan) and a linear multifrequency 12L probe of the ultrasonography device (General Electric VIVID e model, G.E. Medical Systems, Phoenix-USA) after appropriate sterile preparation. Then, while the patient was in the supine position, the needle was inserted through the midclavicular line at the level of the 3rd intercostal space and advanced to the interfascial plane between the pectoralis major and pectoralis minor muscles. After confirming the location of the needle with 2-3 mL of saline, 10 mL of 0.25% bupivacaine were injected. Then, the needle was advanced to the interfascial plane between the pectoralis minor and serratus anterior muscle, and 20 mL of 0.25% bupivacaine were administered with the same procedure. (A total of 20 mL of local anesthetic was administered to patients' weighing less than 40 kg). Block was repeated on the other side using similar technique.

Surgical technique

All surgeries were performed by a single surgeon with at least five years of experience in this technique (SK). The authors prefer to perform MIRPE with the patient in the supine position, by positioning the right and left arm at 90 degrees of abduction. Three-cm transverse incisions are located bilaterally at the anterior axillary line. Incisions are made at the level where the pectus excavatum is most significant. A separate 0.5-cm incision for 30° 5-mm thoracoscope is located at the posterior axillary line. A template is shaped to the desired chest shape, and the bar bent to conform to this template using benders. An introducer is inserted in the contralateral side, and the introducer is taken out from the thoracic cavity through the skin incision. Then, the tape is hooked up to the introducer and pulled back. The bar is inserted into the thoracic cavity with the help of the tape and rotated until it sits retrosternal to correct the deformity. Following the correction, a small catheter is used to drain intrathoracic air from the lung to get it fully expanded.

Postoperative pain management

For multimodal postoperative analgesia, 20 mg of tenoxicam (IV) and 1 g of paracetamol were administered routinely every 12 hours. In addition, demand-only morphine IV PCA was provided to all patients with patient-controlled analgesia (PCA) device (CADD-Legacy PCA Ambulatory Infusion Pump, Model 6300/Smiths Medical/USA). The PCA regimen provided a bolus dose of 0.025 mg/kg, lock time of 20 minutes, and maximum 4-hour dose of 8 mg. No basal infusion was provided. The patients' pain scores at the postoperative 1st, 4th, 8th, 12th, and 24th hours were recorded by the PACU nurse. Postoperative pain was assessed using a numeric rating scale (NRS) (0-10; 0 = no pain and 10 = severe unbearable pain). The static numeric rating scale (S-NRS) was evaluated when patients were at rest, while dynamic NRS (D-NRS) was assessed by asking patients to gently cough.¹⁷ For patients with pain scores ≥ 4 , 0.5 mg/kg of tramadol hydrochloride were administered as rescue analgesia. Patients' cumulative opioid dose, request for rescue analgesia, and additional opioid-related side effects, such as nausea, vomiting, pruritus, and urinary retention, were recorded for the first 24 hours.

Outcome measurements

The primary objective of this study was to determine the total opioid consumption in the PECS-II and control groups 24 hours after surgery. NRS scores, the dose of opioids used as a rescue analgesic, and opioid-related adverse events were used as secondary outcome measures.

Statistical analysis

The G*Power 3.1.9.2 program was used to calculate the study's sample size. By statistical analysis from a previous study in MIRPE¹⁸, the cumulative used opioid doses were 0.29 ± 0.08 mg/kg in the block group and 0.46 ± 0.18 mg/kg in the control group twelve hours after surgery. Therefore, the authors assumed that there should be at least 16 participants in each group, based on an effect size of 1.22, 95% power, and 5% significance level for dropout compensation.

The data collected in the study were evaluated with the SPSS 22.00 program for Windows 10. The Kolmogorov-Smirnov test was used to check the normality of the data distributions. For descriptive statistics, categorical variables are given percentages (%) and numerical variables as mean $\pm$ standard deviation. In comparing the quantitative data of the two groups when the normality conditions were met, the two-sample independent t-test was used. In contrast, Fisher's exact test was used when the variables were qualitative. Mann-Whitney U-test was used for quantitative variable data comparisons when normality conditions were not met. The statistical significance level of alpha was accepted as $p < 0.05$.

Results

The medical records of one hundred seventy-one patients who underwent MIRPE with or without PECS-II block were reviewed. Forty-six patients were excluded from the study due to incomplete data, age, other fascia plane blocks, or different used local anesthetic doses. Accordingly, the data of fifty-eight patients from the PECS group and sixty-seven patients from the control group were analyzed (Figure 1). There was no statistically significant difference between the groups regarding demographic data (Table 1). While the duration of surgery was similar in both groups, the duration of anesthesia was statistically longer in the block group ($p = 0.791$ and $p < 0.001$, respectively) as expected given the practice of performing blocks in pediatric patients after induction of general anesthesia. The opioid bolus dose for IV PCA was calculated based on the patient's weight, and the opioid doses administered to the groups were compared as morphine equivalents per kg. Accordingly, cumulative morphine consumption was significantly lower in all time periods in PECS group ($p < 0.001$) (Table 2, Figure 2). Similarly, the need for postoperative rescue analgesia was significantly lower in the PECS group ($p = 0.012$). Differences in the incidence of postoperative nausea, vomiting, and itching were not statistically significant between the groups ($p = 0.258$, $p = 0.284$, and $p = 0.499$, respectively)

Table 1
Demographic Characteristics and Surgical Data

	Group P (n=58)	Group C (n=67)	p value
Age (yr)	17.55±2.25	17.64±1.78	0.804
Gender			
Male	56 (96.6%)	65 (97.0%)	0.303
Female	2 (3.4%)	2 (3.0%)	
Height (cm)	170.44±4.5	172.04±4.24	0.288
Weight (kg)	57.1±7.52	58.1±5.13	0.394
BMI	19.60±2.0	19.80±1.49	0.536
ASA			
I	54 (93.1%)	64 (95.5%)	0.703
II	4 (6.9%)	3 (4.5%)	
Duration of anesthesia (min)	107.10±20.30	94.±24.29	< 0.001
Duration of surgery (min)	71.25±20.41	72.16±17.75	0.791
Intraoperative remifentanyl use (µg/kg/min)	0.043±0.015	0.045±0.013	0.392
Number of bar (s) (1/2)	58 (100%) / 0	67 (100%) / 0	1.000
Length of stay (day)	3.17±0.38	3.32±0.63	0.095

Data are presented as the mean ± standard deviation and number of patients (%). ASA = American Society of Anesthesiologists, BMI = body mass index.

(Table 2). In the first 12 hours postoperatively, static and dynamic NRS values were significantly lower in the PECS group. However, the static NRS scores at the 24th hour were statistically lower in the PECS group, while dynamic NRS

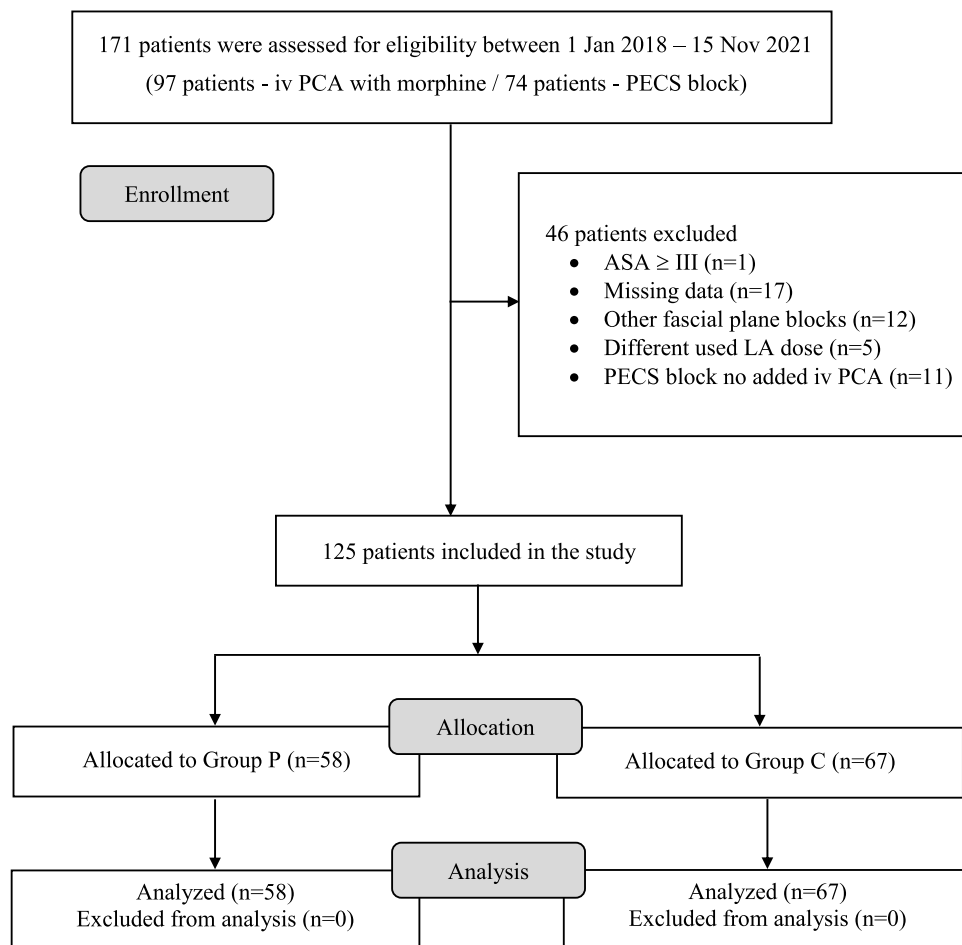


Fig. 1. Flow chart of the study

Table 2
Comparison of Opioid Consumptions and Incidence of Adverse Effects Between Groups

	Group P (n=58)	Group C (n=67)	p value
Morphine consumption (per kg)			
0-1 h	0.025 (0.02-0.07)	0.05 (0.02-0.07)	0.001
0-4 h	0.087 (0.02-0.12)	0.125 (0.07-0.17)	< 0.001
0-8 h	0.200 (0.05-0.35)	0.275 (0.10-0.37)	< 0.001
0-16 h	0.262 (0.10-0.40)	0.375 (0.22-0.55)	< 0.001
0-24 h	0.325 (0.10-0.52)	0.425 (0.22-0.70)	< 0.001
Rescue analgesic requirement (Y/N)	36 / 22	54 / 13	0.012
Vomiting (Y/N)	4 / 54	9 / 58	0.258
Nausea (Y/N)	2 / 56	6 / 61	0.284
Itching (Y/N)	0 / 58	2 / 65	0.499
Constipation (Y/N)	0 / 0	0 / 0	1.000
Sedation/Confusion (Y/N)	0 / 0	0 / 0	1.000
Breathing depression (Y/N)	0 / 0	0 / 0	1.000

Values are presented as the median or number of patients.

values were similar between the two groups ($p = 0.003$ and $p = 0.949$, respectively). When the groups were compared in terms of severe pain (NRS ≥ 7), a significant difference was found

while coughing only at 4 and 8 hours ($p = 0.020$ and $p = 0.034$, respectively) (Table 3).

Discussion

This study indicated reduced opioid consumption and reduced pain scores in patients undergoing minimally invasive repair of pectus excavatum (Nuss procedure) who received PECS-II blocks compared to patients who did not receive blocks. In randomized controlled studies conducted after the definition of PECS-II block, the analgesic efficacy of this block, especially in mastectomy surgery, has been demonstrated.^{14,15,19} To date, different analgesic techniques, such as thoracic epidural analgesia, TPVB, intercostal nerve block, continuous chest wall infusion of local anesthetics, intercostal cryoablation, and intravenous patient-controlled analgesia (PCA), have been used to manage analgesia after MIRPE.^{18,20,21}

This retrospective study was the first in the literature of which the authors are aware to evaluate PECS-II block in the context of MIRPE. Previous reports of interfascial plane blocks for MIRPE have described serratus plane block and erector spinae plane block.^{19,22} These studies reported positive

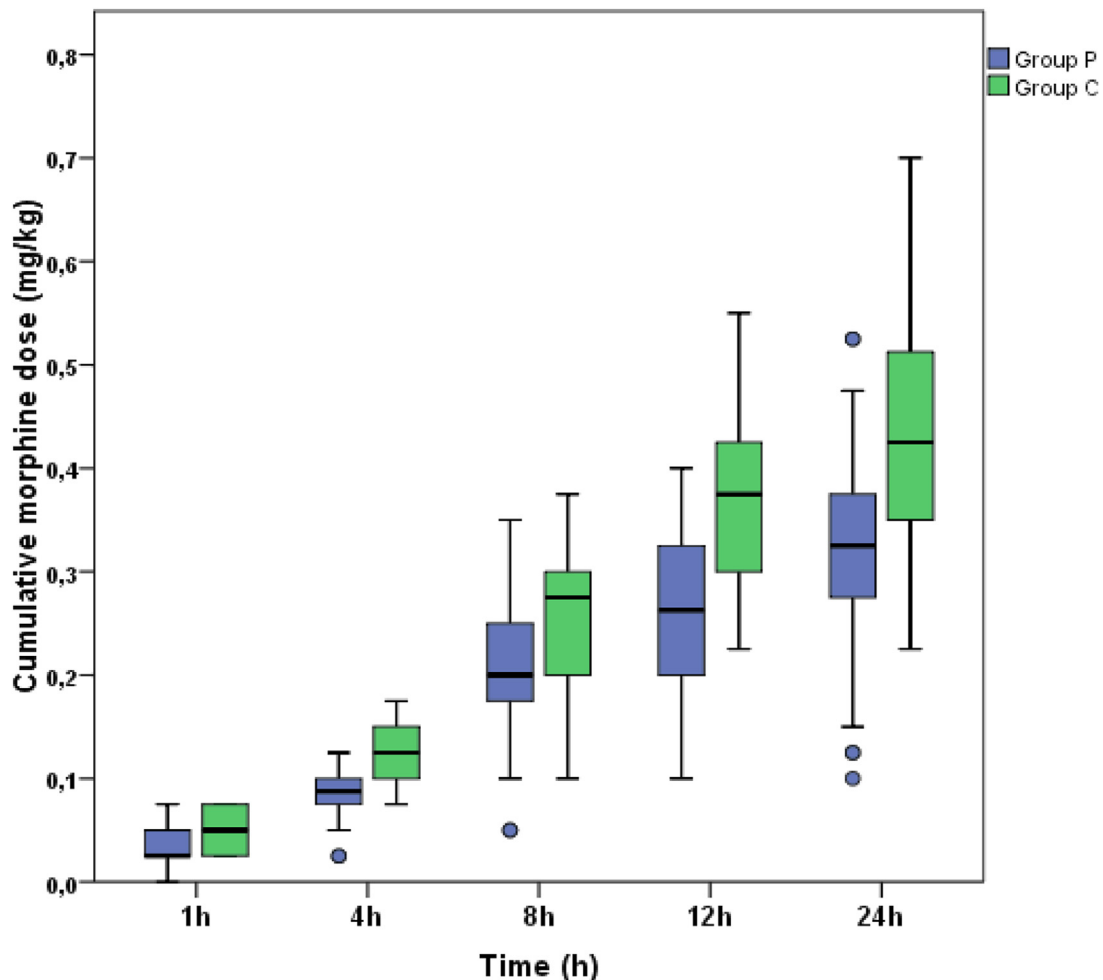


Fig. 2. Postoperative median cumulative morphine doses at the 1-, 4-, 8-, 12- and 24-hour follow-ups.

Table 3
Comparison of NRS Scores Between Groups

		Group P (n=58)	Group C (n=67)	p value
S-NRS	1 h	3 (0-7)	4 (2-8)	0.001
	4 h	2 (0-5)	3 (1-6)	0.002
	8 h	2 (0-5)	3 (1-5)	0.001
	12 h	2 (0-5)	2 (0-5)	0.014
	24 h	1 (0-3)	2 (0-3)	0.003
D-NRS	1 h	3 (1-7)	4 (2-9)	< 0.001
	4 h	2 (0-6)	3 (2-8)	< 0.001
	8 h	2 (0-6)	3 (1-8)	< 0.001
	12 h	2 (0-6)	2 (1-7)	0.001
	24 h	2 (0-5)	2 (0-7)	0.949
Severe pain for rest (S-NRS ≥ 7)	1 h	4 (6.9%)	9 (13.4%)	0.233
	4 h	0 (0%)	0 (0%)	N/A
	8 h	0 (0%)	0 (0%)	N/A
	12 h	0 (0%)	0 (0%)	N/A
	24 h	0 (0%)	0 (0%)	N/A
Severe pain while coughing (D-NRS ≥ 7)	1 h	5 (8.6%)	11 (16.4%)	0.193
	4 h	0 (0%)	6 (9%)	0.020
	8 h	0 (0%)	5 (7.5%)	0.034
	12 h	0 (0%)	1 (1.5%)	1.000
	24 h	0 (0%)	1 (1.5%)	1.000

Data are presented as median or number of patients (%).

S-NRS, static numerical rating scale; D-NRS, dynamic numerical rating scale; h, hour.

results, particularly reduced opioid consumption. The use of the PECS-II block in various thoracic surgical procedures has been described.^{16,23} PECS-II block has been compared with other analgesic techniques (IV PCA, other plane blocks) particularly for mastectomy surgery; advantages such as lower pain scores and less postoperative opioid consumption have been demonstrated.^{14,15,18,24}

It is hypothesized that post-thoracic surgery pain may be caused by crushing of the intercostal nerves during the surgical procedure.²⁵ After thoracic surgery, the severity of pain in children and young adults may be less than in adults because anatomic structures, such as bones, tendons and ligaments, are more flexible. Chronic post-thoracotomy pain, which is defined as pain lasting more than 3 months after surgery, affects 16% of this age group and impairs patients' activities of daily living and sleep quality.^{26,27}

Post-MIRPE pain is caused by a variety of factors, including outward pressure from the Nuss bar placed through the midaxillary line during surgery, incisional pain, muscular spasms, and the release of inflammatory mediators as a result of surgery.^{28,29} While this surgery requires relatively small incisions, intraoperative traction, dissection of the intercostal muscles, and additional stretching and pressing on the chest wall during the procedure cause considerable postoperative pain.²² PECS block can provide pain relief on the chest wall by blocking the third-sixth intercostal nerves and long thoracic

nerve, leading to relaxation of the pectoral muscles.^{14,30} In their meta-analysis,²⁴ authors analyzed thirteen different randomized studies with PECS-II block in patients who had breast cancer surgery, and reported that PECS-II block was more effective in decreasing opioid use or pain scores compared to systemic analgesia.

This study identified lower static pain scores in patients undergoing MIRPE with PECS-II block for the first postoperative 24 hours. However, when the groups were compared in terms of severe pain, no significant difference was found except for the dynamic NRS scores at the 4th and 8th hours. The nerves that innervate the anterior chest wall are mainly lateral and anterior cutaneous branches of the second-sixth thoracic intercostal nerves and branches of the supraclavicular nerves.^{31,32} The sternal area primarily is innervated by the anterior cutaneous branches of the thoracic intercostal nerves, while the lateral cutaneous branches innervate the anterior and middle axillary lines. PECS-II block anesthetizes the lateral cutaneous branches of the thoracic intercostal nerves, the medial and lateral pectoral nerves, and the long thoracic nerves. Thus, the local anesthetic given for the PECS-II block may not be adequate to block the anterior cutaneous branches of the thoracic intercostal nerves.³⁰ Similarly, in some recent studies, it was suggested that PECS-II block was inadequate to cover parasternal and paraspinal areas.³³⁻³⁶

In a phase II study investigating the anesthetic and analgesic effects of the combination of PECS-II and parasternal block in patients undergoing breast cancer surgery, it was reported that 95% of the patients achieved an acceptable sensory block and expected analgesic and anesthetic effects on T2-to-T6 dermatomes.³⁷

In a case report in which a PECS-II block was performed for a patient with multiple bilateral anterolateral costal fracture and sternum body fracture due to anterior chest trauma, PECS-II block provided pain relief for up to 48 hours. The authors attributed this effective analgesia to the fact that local anesthetics spread in all directions along the fascial plane and reached the anterior cutaneous branches.³⁸ Therefore, in order to provide more effective analgesia around the sternum, PECS-II block could be combined with a parasternal block that numbs the anterior cutaneous intercostal nerves.^{37,38}

The PECS-II block is an easy-to-apply superficial block. It can be performed safely in a short time in the supine position. Complications, such as intravascular injection (thoracoacromial artery and cephalic vein) or pneumothorax, may occur. As in other reports, none of the patients in this study experienced such complications. Although encouraging, the small cohort size prevented meaningful assessment of safety.

The study had other limitations. The most important limitation of retrospective analysis was its ability to identify associations without being able to assess causality. Furthermore, since it was a retrospective study, randomization could not be performed, and the patients were included in the study consecutively according to the order of admission. In addition, the results of this study may not be generalizable to patients treated in healthcare centers with different dosing protocols. In addition, dermatomal evaluation could not be performed

because the blocks were performed after induction of anesthesia. Still, the spread of local anesthetic to the correct area was confirmed by ultrasound guidance.

Conclusion

In this retrospective analysis, PECS-II block was associated with reduced pain scores at rest for up to 12 hours, and with decreased opioid consumption for up to 24 hours, following minimally invasive repair of pectus excavatum (Nuss procedure).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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